# Anh (Joe) Nguyen

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## Selected Research Experience

**Fujitsu Research of America — Research Intern** Sep 2026 - Dec 2026

– Computer-use LLM agents: Build a self-evolving (test-time adapation) world model for computer-use agents in enterprise environments.

**Oregon State University — Research Asisstant** Jan 2022 - present

– Self-evolving LLM agents: Contributed to a training-free method for LLM agents to self-improve on open-ended tasks via evolving feedback, without any parameter updates. [2]

– LLM-based Computer use agent: Propose a simple yet effective method (RAG-PJ) to use multimodal tutorials for computer-use agents. We show that RAG-PJ achieves similar performance to other baselines *without using expensive world model rollouts*. [1]

– Language model-based RL agent: Developed an agent that can understand language descriptions of environment dynamics and act accordingly. The agent achieves *state-of-the-art policy generalization* over new language and unseen environment dynamics, increasing 6%-21% success rates over existing baselines (Dynalang, EMMA, LWM) [3]

– Visual language referring expression: Collected a new dataset in referring expressions for underwater objects. Compared and analyzed results from the state-of-the-art methods OFA and CLIP: both have low accuracy performance on underwater objects that are rare in territorial settings.

**Singapore Management University — Research Engineer** Nov 2019-Nov 2021

– Developed a method that can clarify ambiguous language queries in visual domains (referring expression): detect the most ambiguous portion of the query based on neural module network, then raise an relevant clarifying question to users.

– The method results in more efficient clarifying process: less than 10% of clarifying questions than existing baselines in CLEVR-Ref and Ref-Reasoning.

## Publication

1. Anh Nguyen, Stefan Lee. Look It Up Before Looking Ahead: Tutorials Reduce the Benefit of World Models in Computer-Use Agents, submitted to NAACL 2027.

2. Tianyi Xu, Yaolun Zhang, Yiran Wu, Anh (Joe) Nguyen, Shengyu Dai, Saba Moeinizade, Zhenwen Shao, Qingyun Wu, Huazheng Wang. Training-Free Agent Self-Improvement on Open-Ended Tasks via Evolving Feedback, submitted to ICLR 2027.

3. Anh Nguyen, Stefan Lee. Language-conditioned world model improves policy generalization by reading environmental descriptions, accepted at Bridging Language, Agent, and World Models workshop at NeurIPS2025.

4. Anh Nguyen, Duy Tue Tran Van, Minh Quoc Nghiem. Bachelor Thesis: Extractive summarization with bidirectional encoder representations from transformers: achieves state-of-the-art *long-text* summarization on CNN/Daily Mail (2019).

5. Duy Phung, Tu Minh, Anh Nguyen, Tien Dinh, DTA Hunter System: A new statistic-based framework of predicting future demand for taxi drivers, accepted for presentation @ SoICT 2017 (The

Eighth International Symposium on Information and Communication Technology)

## Education

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| 2022 - 2027 (Expect) | Msc (thesis) and PhD at **Oregon State University**. GPA: 3.9/4.0<br>Advised by Professor Stefan Lee and Professor Huazheng Wang |
| 2014 - 2019 | Bachelor's Degree at **VNU-Ho Chi Minh University of Science**<br>(GPA: 3.8/4.0, top 5% of Department)<br>Advised by Dr. Nghiem Quoc Minh |

## Academic Projects

- **Language-conditioned LLM-based world model**: Identified the research gap of current LLM-based model based RL systems: they failed to incoperate language into the world model, thus unable to change world modeling on the fly based on language.
- **Exploration in RL**: Compared different exploration strategies in model-based RL: count-based, curiosity-based and Monte-Carlo dropout in PointMaze
- **Search in Games (Sudoku and (M,N,K))**: Compared different tree search in Sodoku and (M,N,K) game: Monte-Carlo Tree Search, A* and Minimax with Alpha-Beta prunning.

## Awards

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| 2025 | Scholarly Presentation Award from Oregon State University. |
| 2022 | Vietnam Education Foundation (VEF) 2.0 candidate for Ph.D programs in the USA. |
| 2019 | Fully funded exchange at APEC Digital Innovation and Entrepreneurship Forum (Taiwan) |
| 2019 | Fully funded exchange at GKS ASEAN SCIENCE at Kyungwoon University (Korea) |
| 2019 | Fully funded exchange at KAIST (South Korea) |
| 2019 | Fully funded exchange at Southeast Asia Machine Learning School |
| 2019 | Scholarship from Shinhan Bank for excellent students |
| 2017 | Scholarship from American Chamber of Commerce Vietnam for excellent students |
| 2017 | Scholarship from Global Cybersoft for excellent students |
| 2017 | Scholarship from Esilicon Vietnam for excellent students |
| 2016 | ACM-ICPC Vietnam National Competitive programming: Third Prize |
| 2016 | HCMc University of Science Olympiad in Informatics : Honorable mention |
| 2014 | Viethope Scholarship for excellent college students |
| 2013 | National Vietnam Olympiad in Informatics: Honorable Mention |
| 2013 | Vallet scholarship for excellent students |

## Services

- Reviewer: NeurIPS 2027, NAACL 2027, EMNLP 2026

## Skills

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| Programming | C++ (expert), Python (expert), Java (familiar with), competitive programming |
| Research (expert) | LLM training: RL training |